

Trainings ▾

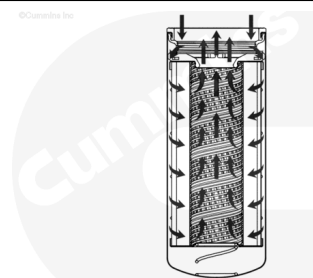
General Information

The recommended oil filter for this engine is a combination NanoNet® oil filter. This filter contains 2/3rd section of StrataPore® media and 1/3rd section of high efficiency NanoNet® media.

The combination media design has been optimized to distribute oil flow during normal engine operation; however, during extreme engine oil conditions (additive dropout, contamination, soot etc.)

Oil circulates through the StrataPore® full-flow portion of the filter, into the oil cooler housing, and is directed to the engine oil rifles.

A smaller portion of the oil circulates through the lower bypass NanoNet® full flow portion of the filter and is directed to the engine oil rifles.



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Lubricating Oil

Use Valvoline Premium Blue™ 15W-40 multiviscosity oil, or its equivalent, which meets CES 20081 and CES 20078 oil classifications for turbocharged engines.

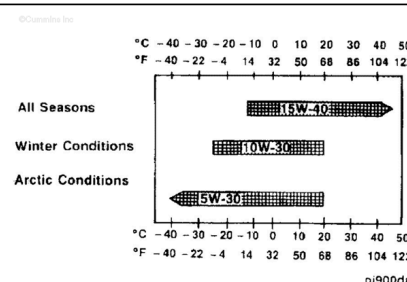
Refer to Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810340.html>), for additional oil information.



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⚠ CAUTION ⚠

Limited use of oils such as 5W-30 or 10W-30 can aid in starting the engine and providing sufficient oil flow at ambient temperatures below -5°C [23°F]. However, the continuous use of low-viscosity oils can decrease engine life.



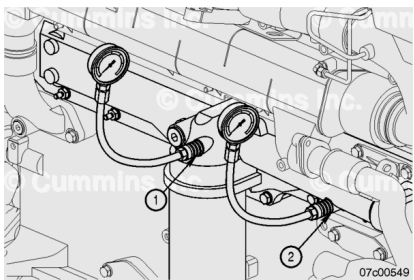
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See the accompanying chart to determine the best weight of oil for the engine operating conditions.

Measure

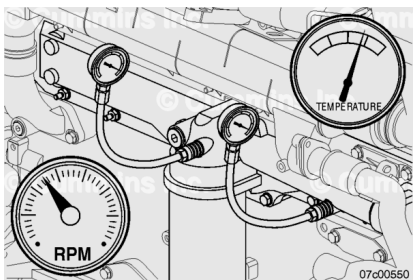
Remove the oil plugs from the lubricating oil cooler housing at the filter inlet (2) and outlet (1) pressure ports and install manual gauges.

The following parts, or equivalent, are available for use:



Part Number	Description
3824844	Compuchek™ fitting for port size M14x1.5
3376920	Compuchek™ coupling (1/4-inch pipe thread)
3164491	Electronic pressure adapter for multimeter (1/4-NPT pipe)
3164488 or 3164489	Electronic digital multimeter

Start and operate the engine until the oil temperature reaches or exceeds 70°C [160°F].



Engine Information	
Oil filter type:	
Miles on filter:	
Oil type:	

Operate the engine at each rpm called for and record the corresponding pressure values:

Oil Temperature	
Start of test:	
End of Test	

	Engine RPM	Oil Pressure Filter Inlet	Oil Pressure Filter Outlet	Inlet - Outlet = Differential Pressure	INSITE™ electronic service tool
Low Idle					
High Idle					

A pressure drop greater than 172 kPa [25 psi], at operating temperature using 15W-40 oil, indicates the filter is plugged.

Identify the causes of a plugged filter. Verify the Cummins Inc. maintenance guidelines are being met.

Refer to Procedure For possible fluid contamination, 007-083 (Lubricating Oil Filter and Filter Analysis) in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-083.html>)

Change both the lubricating oil and lubricating oil filter, if plugged.

Reference symptom tree t103-56 (Lubricating Oil Filter Plugged) in Section TS.

For additional information about lubricating oil filter plugging, refer to Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810340.html>).

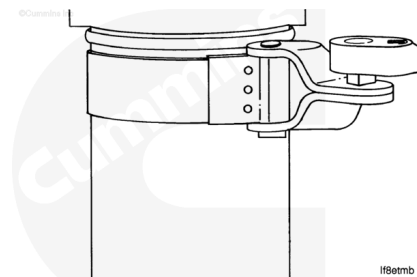
Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Note : If the lubricating oil filter head adapter comes off during the removal of the lubricating oil filter, then reinstall the oil filter head adapter. Use a small amount of Loctite™ on the threads of the oil filter head adapter and tighten. Refer to Procedure 007-018 (Lubricating Oil Filter Head Adaptor) in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-018.html>)

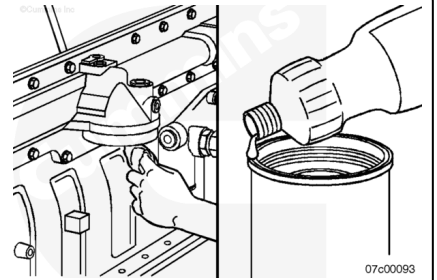
Use oil filter wrench, Part Number 3375049, 3397929, or 3400157 to remove the oil filter.

Install

Note : The o-ring can stick onto the filter head. Make sure it is removed before installing a new filter.

Clean the oil filter head surface.

Use clean engine oil to lubricate the gasket surface of the filter.



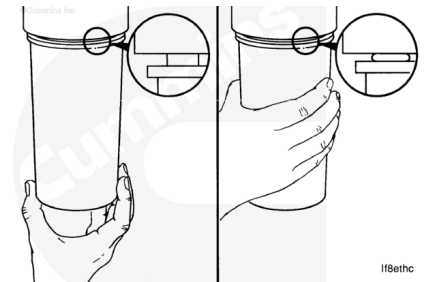
⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal.

Fill the oil filter with clean 15W-40 oil.

Install the oil filter onto the oil filter head. Tighten the filter until the gasket contacts the filter head surface.

Use oil filter wrench, Part Number 3375049, 3397929, or 3400157 to tighten the filter an additional 3/4 to 1 turn, or follow the instructions supplied with the filter.



Confirm that the oil level in the pan is to the full mark.

Operate the engine until the water temperature reaches 82°C [180°F].

Check the oil pressure and for leaks.

